Lecture-2

Register Transfer

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Instruction to Register

- **Register** is a very fast computer memory, used to store data/instruction in-execution.
- A register is a group of flip-flops with each flip-flop capable of storing one bit of information.
- An n-bit register has a group of n flip-flops and is capable of storing binary information of n-bits.
- The registers used by the CPU are often termed **Processor registers**

Instruction to Register (Contd…)

- A **Processor register** may hold an instruction, a storage address, or any data.
- The computer needs processor registers for **manipulating data** and a register for **holding a memory address.**
- The resister holding the memory location is used to calculate the address of the next instruction after the execution of the current instruction is completed.

Instruction to Register

Some of the commonly used registers are:

- **Accumulator:** This is the common register, used to store data taken out from the memory
- **General Purpose Registers:** This is used to store data intermediate results during program execution. It can be accessed via assembly programming.
- **Special Purpose Registers:** Users do not access these registers. These registers are for computer systems.

Instruction to Register

Some of the commonly used registers are:

- **Special Purpose Registers:** Users do not access these registers. These registers are for computer systems.
  - **MAR:** Memory Address Registers either store the memory address from which data will be fetched to the CPU registers or the address to which data will be sent and stored via the system bus.
  - **MBR:** Memory Buffer Register stores instruction and data received from the memory and sent from the memory.
  - **PC:** Program Counter points to the next instruction to be executed.
  - **IR:** Instruction Register holds the instruction to be executed.

Microoperations

- The operations on the data in registers are called micro operations.
- The functions built into registers are examples of micro operations
  - Shift
  - Load
  - Clear
  - Increment
  - …

Organization Of A Digital System

- **Definition of the (internal) organization of a computer**

  - Set of registers and their functions

  - Micro-operations set

    Set of allowable microoperations provided by the organization of the computer

  - Control signals that initiate the sequence of microoperations (to perform the functions)

Register Transfer Level

- Viewing a computer, or any digital system, in this way is called the register transfer level

- This is because we're focusing on
  - The system's registers
  - The data transformations in them, and
  - The data transfers between them.

Register Transfer Language

- Rather than specifying a digital system in words, a specific notation is used, *register transfer language*
- For any function of the computer, the register transfer language can be used to describe the (sequence of) microoperations
- Register transfer language
  - A symbolic language
  - A convenient tool for describing the internal organization of digital computers
  - Can also be used to facilitate the design process of digital systems.

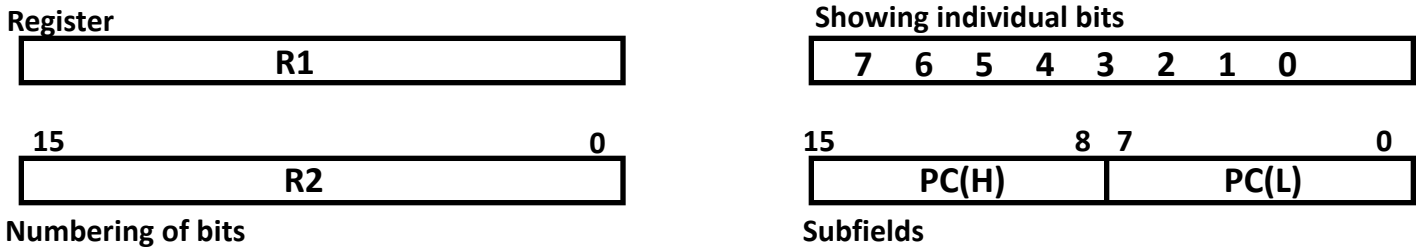
Designation Of Registers

- Registers are designated by capital letters, sometimes followed by numbers (e.g., A, R13, IR)
- Often the names indicate function:
  - MAR - memory address register
  - PC - program counter
  - IR - instruction register
- Registers and their contents can be viewed and represented in *various ways*
  - A register can be viewed as a single entity:
  - Registers may also be represented showing the bits of data they contain

Designation Of Registers

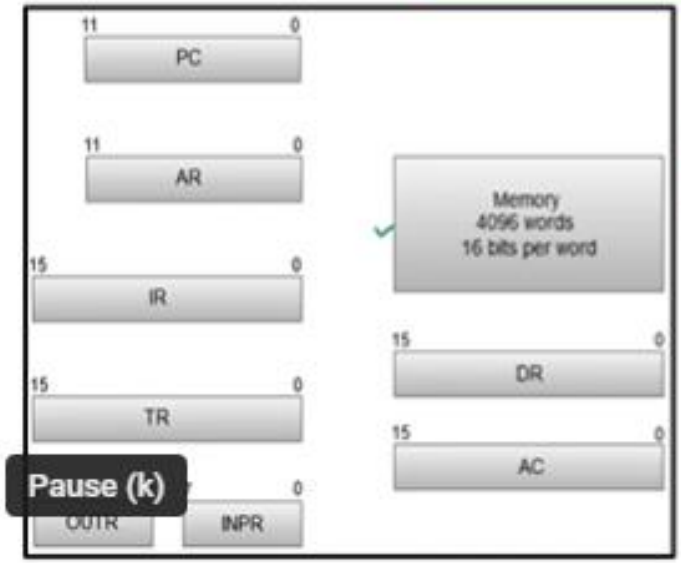
- Designation of a register
  - a register
  - portion of a register
  - a bit of a register

- Common ways of drawing the block diagram of a register
  - (0-7) low order bit PC(L) and (8-15) high order bit PC(H)



Register Transfer

- The computer needs a processor for manipulating data and a register for holding a memory address



Register	Symbol	Number of bits	Function
Data register	DR	16	Holds memory operand
Accumulator	AC	16	Processor register
Program counter	PC	12	Holds address of the instruction
Address register	AR	12	Holds address for the memory
Instruction register	IR	16	Holds instruction code
Temporary register	TR	16	Holds temporary data
Input register	INPR	8	Carries input character
Output register	OUTR	8	Carries output character

Register Transfer

- Copying the contents of one register to another is a register transfer
- A register transfer is indicated as : R1 ← R2

  – In this case the contents of register R2 are copied (loaded) into register R1
  – A simultaneous transfer of all bits from the source R2 to the destination register R1, during one clock pulse
  – Note that this is a non-destructive; i.e. the contents of R2 are not altered by copying (loading) them to R1

Register Transfer

- A register transfer such as R3 ← R5 implies that the digital system has
  - the data lines from the source register (R5) to the destination register (R3)
  - Parallel load in the destination register (R3)
  - Control lines to perform the action

Control Functions

- Often actions need to only occur if a certain condition is true
- This is similar to an "if" statement in a programming language
- In digital systems, this is often done via a *control signal*, called a *control function*
  - If the signal is 1, the action takes place
- This is represented as- P: $R2 \leftarrow R1$

  Which means "if P = 1, then load the contents of register R1 into register R2",

  i.e., if (P = 1) then ($R2 \leftarrow R1$)

Simultaneous Operations

- If two or more operations are to occur simultaneously, they are separated with commas

$$P: \ R3 \leftarrow R5, MAR \leftarrow IR$$

- Here, if the control function P = 1, load the contents of R5 into R3, and at the same time (clock), load the contents of register IR into register MAR

Basic Symbols For Register Transfers

Symbols	Description	Examples
Capital letters & numerals	Denotes a register	MAR, R2
Parentheses ()	Denotes a part of a register	R2(0-7), R2(L)
Arrow ←	Denotes transfer of information	R2 ← R1
Colon :	Denotes termination of control function	P:
Comma ,	Separates two micro-operations	A ← B, B ← A

Summary Of Register Transfer Microoperations

A ← B	Transfer content of reg. B into reg. A
AR ← DR(AD)	Transfer content of AD portion of reg. DR into reg. AR
A ← constant	Transfer a binary constant into reg. A
ABUS ← R1,	Transfer content of R1 into bus A and, at the same time,
R2 ← ABUS	Transfer content of bus A into R2
AR	Address register
DR	Data register
M[R]	Memory word specified by reg. R
M	Equivalent to M[AR]
DR ← M	Memory *read* operation: transfers content of memory word specified by AR into DR
M ← DR	Memory *write* operation: transfers content of DR into memory word specified by AR